

Bellabeat Smart Device Usage Analysis

Google Data Analytics Professional Certificate Capstone Project

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Tools Used

- Google Sheets
- Google BigQuery (SQL)
- Tableau Public

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1. ASK PHASE

Business Task

Bellabeat is a high-tech company that manufactures health-focused smart products designed to help users monitor and improve their overall wellness. The objective of this analysis is to examine Fitbit smart device usage data to identify consumer usage patterns and generate insights that can inform Bellabeat's marketing strategy. The findings will be used to provide data-driven recommendations that help Bellabeat better understand its target customers and encourage greater adoption of its smart wellness products.

Stakeholders

The key stakeholders for this project are:

- **Urška Sršen** – Bellabeat co-founder and Chief Creative Officer.
 - **Sando Mur** – Bellabeat co-founder and key member of the executive team.
 - **Bellabeat Marketing Analytics Team** – Responsible for analyzing consumer behavior and developing marketing strategies based on data insights.
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Business Objectives

The analysis seeks to answer the following questions:

1. How active are Fitbit users on average?
2. Which days of the week are users most active?
3. How much sleep do Fitbit users get on average?
4. Do users who take more steps burn more calories?
5. Do users with higher daily activity levels tend to sleep longer?

The answers to these questions will be used to identify consumer usage patterns and develop marketing recommendations for Bellabeat.

2. PREPARE PHASE

Data Source

The dataset used for this analysis is the **Fitbit Fitness Tracker Data**, obtained from Kaggle. The dataset was made available by **Mobius** under the **CC0: Public Domain** license. It contains personal fitness tracker data collected from Fitbit users through a distributed survey conducted using Amazon Mechanical Turk between **March 12, 2016, and May 12, 2016**.

The dataset includes daily activity, sleep, heart rate, and physical activity metrics that can be used to analyze smart device usage patterns and support Bellabeat's marketing strategy.

Data Credibility

The credibility of the dataset was evaluated using the **ROCCC** framework.

Reliability

The dataset contains activity records collected from **real Fitbit users**, making it suitable for analyzing actual smart device usage behavior. However, the sample consists of **33 unique users**, which is relatively small and limits the extent to which the findings can be generalized to a broader population.

Originality

The dataset was made available through Kaggle by Mobius and is not the original source of data collection. It was collected through Amazon Mechanical Turk and shared under a public domain license.

Comprehensiveness

The dataset contains daily activity, sleep, distance, and calorie information that is sufficient for analyzing smart device usage patterns. However, it does not include demographic information such as age, gender, or location, limiting the ability to analyze specific user groups.

Current

The data was collected in **2016** and may not accurately reflect current smart device usage patterns or consumer behavior.

Cited

The dataset includes information about its source, collection method, and licensing, making it appropriately documented for analytical use.

Data Integrity

During the initial review, the case study documentation stated that the dataset contained data from **30 Fitbit users**. However, analysis of the unique user IDs showed that the dataset actually contains records for **33 unique users**. This inconsistency was documented as a limitation because it represents a discrepancy between the documentation and the dataset, although it does not invalidate the data.

Limitations

The following limitations should be considered when interpreting the results:

- The dataset contains records for **33 unique Fitbit users**, representing a relatively small sample size that may not reflect the broader population.
- The dataset does not include demographic information such as age, gender, or location. Since Bellabeat primarily targets women, this limits the ability to determine whether the findings accurately represent Bellabeat's target market.
- The data was collected over approximately one month in **2016** and may not reflect current smart device usage patterns.
- During data validation, some records contained inconsistencies, such as **1,440 sedentary minutes (24 hours)** while also recording physical activity or calorie expenditure. Because the dataset documentation does not explain these values, the records were retained and documented as a limitation rather than removed.

3. PROCESS PHASE

Data Preparation

Before analysis, the selected datasets were cleaned and validated to ensure accuracy, consistency, and reliability. **Google Sheets** was used to inspect the data, identify quality issues, perform transformations, and prepare the datasets for SQL analysis. **Google BigQuery** was then used to perform the data analysis because it provides efficient querying, aggregation, and exploration of large datasets.

Selected Datasets

The following datasets were selected because they directly support the business task:

DailyActivity

Contains daily activity metrics including:

- Daily steps
- Calories burned
- Distance traveled
- Activity intensity levels

This dataset was used to analyze users' physical activity patterns.

SleepDay

Contains daily sleep metrics including:

- Total minutes asleep
- Total time in bed

This dataset was used to analyze users' sleeping habits and examine the relationship between physical activity and sleep duration.

Data Validation

To ensure the datasets were suitable for analysis, the following validation checks were performed:

- Checked for duplicate records.
- Checked for missing values.
- Reviewed data types and formatting.
- Verified that each row represented one user's activity for one day.
- Confirmed that the variables were appropriate for answering the business questions.
- Compared the documented number of users with the actual number of unique user IDs.

During validation, it was confirmed that the datasets contained records for **33 unique users**, rather than the **30 users** stated in the accompanying documentation. This discrepancy was documented as a limitation of the dataset.

Data Cleaning and Transformation

The following cleaning and preparation steps were completed before analysis:

- The **DailyActivity** dataset was checked for duplicate records, and no duplicates were found.
 - Three duplicate records were identified and removed from the **SleepDay** dataset, reducing the total number of records from **413** to **410**.
 - Both datasets were checked for missing values, and no missing values were identified.
 - Numeric columns were formatted for consistency and readability without altering the underlying data.
 - Column names were standardized using lowercase letters and underscores to improve readability and simplify SQL queries.
 - A new **Day_of_Week** column was created from the activity date to support weekday trend analysis.
 - Records containing **1,440 sedentary minutes** were investigated. Since there was insufficient evidence to conclude that these records were incorrect, they were retained and documented as a limitation instead of being removed.
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Data Readiness

After completing the cleaning process, the datasets were reviewed again to confirm that:

- Duplicate records had been removed.
- No missing values were present.
- Date and numeric fields were correctly formatted.

- Column names were standardized.
- The datasets were suitable for SQL analysis.

The cleaned datasets were then imported into **Google BigQuery**, where SQL was used to perform exploratory analysis, calculate summary statistics, identify trends, and answer the business questions.

4. ANALYZE PHASE

Overview

After cleaning and validating the data, the datasets were imported into **Google BigQuery** for analysis. SQL was used to calculate summary statistics, identify activity and sleep trends, and explore relationships between physical activity, calorie expenditure, and sleep duration. The analysis addressed the five business questions defined in the Ask phase.

Business Question 1

How active are Fitbit users on average?

Objective

Determine the average daily activity levels of Fitbit users, including steps, distance traveled, calories burned, activity intensity, and sedentary time.

SQL Analysis

```
SELECT  
  
    ROUND(AVG(total_steps), 0) AS average_steps,  
  
    ROUND(AVG(total_distance), 2) AS average_distance,  
  
    ROUND(AVG(calories), 0) AS average_calories,  
  
    ROUND(AVG(very_active_minutes), 0) AS average_very_active_minutes,  
  
    ROUND(AVG(fairly_active_minutes), 0) AS average_fairly_active_minutes,  
  
    ROUND(AVG(lightly_active_minutes), 0) AS average_lightly_active_minutes,  
  
    ROUND(AVG(sedentary_minutes), 0) AS average_sedentary_minutes  
  
FROM `capstone250.Fitbit.daily_activity`;
```

Results

Metric	Average
Daily Steps	7,638
Distance Traveled (miles)	5.49
Calories Burned	2,304
Very Active Minutes	21
Fairly Active Minutes	14
Lightly Active Minutes	193
Sedentary Minutes	991

Business Insight

Fitbit users averaged **7,638 daily steps**, indicating a moderate level of physical activity. Users also traveled an average of **5.49 miles** and burned approximately **2,304 calories** per day.

On average, users spent **21 minutes** engaged in very active exercise, **14 minutes** in fairly active activities, and **193 minutes** in light physical activity. They also recorded an average of **991 sedentary minutes** per day. Because the dataset contains records with **1,440 sedentary minutes** alongside other activity measurements, this value should be interpreted with caution. Nevertheless, the results suggest there is an opportunity for Bellabeat to encourage users to reduce prolonged inactivity through personalized reminders and movement goals.

Business Question 2

Which days of the week are users most active?

Objective

Determine which days of the week Fitbit users recorded the highest average daily step counts and calorie expenditure.

SQL Analysis

```
SELECT
    day_of_week,
    ROUND(AVG(total_steps), 0) AS average_steps,
    ROUND(AVG(calories), 0) AS average_calories
FROM `capstone250.Fitbit.daily_activity`
GROUP BY day_of_week
ORDER BY average_steps DESC;
```

Results

Day of Week	Average Steps	Average Calories
Saturday	8,153	2,355
Tuesday	8,125	2,356
Monday	7,781	2,324

Wednesday	7,559	2,303
Friday	7,448	2,332
Thursday	7,406	2,200
Sunday	6,933	2,263

Business Insight

Fitbit users were most active on **Saturday**, averaging **8,153 steps** per day, closely followed by **Tuesday** with **8,125 steps**. **Sunday** recorded the lowest average daily step count (**6,933 steps**), suggesting that users were generally less active at the end of the weekend.

Average calorie expenditure remained relatively consistent throughout the week and generally followed the same pattern as daily step counts. These findings suggest that Bellabeat could improve user engagement by introducing personalized reminders, activity challenges, or motivational notifications on lower-activity days, particularly Sundays, to encourage more consistent physical activity throughout the week.

Business Question 3

How much sleep do Fitbit users get on average?

Objective

Determine the average amount of sleep Fitbit users get each night and compare it with the total time spent in bed.

SQL Analysis

```
SELECT
```

```
    ROUND(AVG(total_minutes_asleep), 0) AS average_minutes_asleep,
```

```

ROUND(AVG(total_time_in_bed), 0) AS average_minutes_in_bed,

ROUND(AVG(total_time_in_bed - total_minutes_asleep), 0) AS
average_minutes_awake_in_bed

FROM `capstone250.Fitbit.sleep_day`;

```

Results

Metric	Average
Minutes Asleep	419
Minutes in Bed	458
Minutes Awake in Bed	39

Business Insight

Fitbit users slept an average of **419 minutes (approximately seven hours)** per night while spending **458 minutes** in bed. On average, users remained awake for approximately **39 minutes** before falling asleep or after waking.

These findings suggest that Bellabeat has an opportunity to provide users with deeper sleep insights, such as sleep efficiency and personalized sleep recommendations, rather than reporting only total sleep duration.

Business Question 4

Do users who take more steps burn more calories?

Objective

Examine whether higher daily step counts are associated with increased calorie expenditure.

SQL Analysis

```
SELECT
    CASE
        WHEN total_steps < 5000 THEN 'Low Activity'
        WHEN total_steps BETWEEN 5000 AND 9999 THEN 'Moderate Activity'
        ELSE 'High Activity'
    END AS activity_level,
    COUNT(*) AS number_of_days,
    ROUND(AVG(total_steps),0) AS average_steps,
    ROUND(AVG(calories),0) AS average_calories
FROM `capstone250.Fitbit.daily_activity`
GROUP BY activity_level
ORDER BY average_steps;
```

Results

Activity Level	Number of Days	Average Steps	Average Calories
Low Activity	303	2,128	1,807
Moderate Activity	334	7,466	2,355
High Activity	303	13,337	2,744

Business Insight

Users classified in the **High Activity** group averaged **13,337 daily steps** and burned approximately **2,744 calories**, while users in the **Low Activity** group averaged **2,128 steps** and burned approximately **1,807 calories**.

This demonstrates a clear positive relationship between daily step count and calorie expenditure, confirming that daily steps were a major contributor to calories burned within this dataset. Bellabeat can use this insight to motivate users by promoting personalized step goals and progress tracking.

Business Question 5

Do users with higher daily activity levels tend to sleep longer?

Objective

Investigate whether users with higher daily activity levels also record longer sleep durations.

SQL Analysis

```
SELECT
```

```
    CASE
```

```
        WHEN d.total_steps < 5000 THEN 'Low Activity'
```

```
        WHEN d.total_steps BETWEEN 5000 AND 9999 THEN 'Moderate Activity'
```

```
        ELSE 'High Activity'
```

```
    END AS activity_level,
```

```
    COUNT(*) AS number_of_days,
```

```
    ROUND(AVG(d.total_steps),0) AS average_steps,
```

```
    ROUND(AVG(s.total_minutes_asleep),0) AS average_minutes_asleep
```

```
FROM `capstone250.Fitbit.daily_activity` AS d
```

```
INNER JOIN `capstone250.Fitbit.sleep_day` AS s
```

```
ON d.id = s.id
```

```
AND d.activity_date = DATE(s.sleep_day)
```

```
GROUP BY activity_level
```

```
ORDER BY average_steps;
```

Results

Activity Level	Number of Days	Average Steps	Average Minutes Asleep
Low Activity	96	3,054	454
Moderate Activity	149	7,591	422
High Activity	165	12,526	396

Business Insight

Users classified in the **High Activity** group averaged **12,526 daily steps** but slept approximately **396 minutes (6 hours and 36 minutes)** per night. In comparison, users in the **Low Activity** group averaged **3,054 steps** and slept approximately **454 minutes (7 hours and 34 minutes)**.

Within this dataset, higher daily activity levels were associated with shorter average sleep duration. However, this finding should not be interpreted as evidence that increased physical activity causes reduced sleep, as the dataset does not include other factors such as age, occupation, health status, or lifestyle that could influence sleep patterns. This suggests that Bellabeat should encourage users to monitor both physical activity and sleep as complementary aspects of overall wellness.

Summary of Findings

The analysis showed that Fitbit users maintained moderate levels of daily physical activity, averaging **7,638 steps**, traveling **5.49 miles**, and burning approximately **2,304 calories** per day. Activity levels varied throughout the week, with users being most active on **Saturdays** and **Tuesdays**. Sleep analysis revealed that users averaged approximately **seven hours of sleep** per night while spending about **39 minutes awake in bed**.

Further analysis confirmed a strong positive relationship between daily step count and calorie expenditure, indicating that users who recorded more daily steps generally burned more calories. However, users with higher activity levels did not record longer sleep durations, suggesting that physical activity alone does not necessarily correspond to increased sleep within this dataset.

Overall, these findings demonstrate that Fitbit users actively monitor both their physical activity and sleep using smart devices. The insights generated from this analysis provide a strong foundation for developing data-driven marketing recommendations that can help Bellabeat better engage users and promote healthier lifestyle habits.

5. SHARE PHASE

Objective

The objective of the Share phase was to communicate the results of the analysis in a clear, concise, and visually engaging manner. An interactive dashboard was created in **Tableau Public** to present the key findings and support data-driven decision-making for Bellabeat's marketing team.

Data Visualization

The dashboard was designed to answer the five business questions identified during the Ask phase using clear and accessible visualizations.

The dashboard includes the following visualizations:

1. Average Daily Steps by Day of Week

Chart Type: Bar Chart

This visualization compares the average daily step count across the days of the week, making it easy to identify when users are most and least active.

Key Insight:

Saturday recorded the highest average daily step count, while Sunday recorded the lowest.

2. Average Calories Burned by Activity Level

Chart Type: Bar Chart

This visualization compares calorie expenditure across low, moderate, and high activity groups.

Key Insight:

Users with higher daily step counts consistently burned more calories, confirming a strong positive relationship between physical activity and calorie expenditure.

3. Average Sleep Duration by Activity Level

Chart Type: Bar Chart

This visualization compares average sleep duration across different activity levels.

Key Insight:

Higher activity levels were associated with shorter average sleep durations within this dataset.

4. Dashboard Summary

A summary section was included to highlight the most important findings from the analysis, allowing stakeholders to quickly understand the key insights without examining each chart individually.

Dashboard Design Considerations

The dashboard was designed using data visualization best practices to improve readability and accessibility.

These principles included:

- Consistent color schemes across all charts.
 - Clear titles and descriptive axis labels.
 - Minimal visual clutter.
 - Appropriate spacing between visualizations.
 - Easy comparison of values using bar charts.
 - A logical layout that guides readers through the findings.
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Key Findings Communicated

The dashboard communicates the following insights:

- Fitbit users averaged **7,638 daily steps** and **2,304 calories burned** per day.
- Users were most active on **Saturdays** and **Tuesdays**.
- Users averaged approximately **seven hours of sleep** each night.
- Higher daily step counts were associated with greater calorie expenditure.
- Higher activity levels were associated with shorter average sleep duration within this dataset.

These visualizations provide stakeholders with a clear understanding of Fitbit user behavior and serve as the foundation for the recommendations presented in the Act phase.

6. ACT PHASE

Recommendation 1: Encourage Consistent Daily Activity

Supporting Finding

The analysis showed that users were most active on **Saturdays** and **Tuesdays**, while **Sunday** recorded the lowest average daily step count.

Recommendation

Bellabeat should introduce personalized activity reminders, weekly step challenges, and motivational notifications on lower-activity days, particularly Sundays, to encourage users to maintain consistent physical activity throughout the week.

Expected Business Value

Increasing user engagement on less active days can promote healthier habits while encouraging more frequent interaction with Bellabeat's wellness platform.

Recommendation 2: Promote Personalized Step Goals

Supporting Finding

Users with higher daily step counts consistently burned more calories than users with lower activity levels.

Recommendation

Bellabeat should provide personalized daily step goals that adapt to individual users' activity history rather than relying on a single goal for everyone. Progress tracking, milestone badges, and achievement notifications can further motivate users to remain active.

Expected Business Value

Personalized goals can increase user motivation, improve product engagement, and reinforce Bellabeat's value as a wellness companion.

Recommendation 3: Emphasize Balanced Wellness

Supporting Finding

Users with higher activity levels did not record longer sleep durations. This suggests that physical activity alone does not provide a complete picture of overall wellness.

Recommendation

Bellabeat should encourage users to monitor both physical activity and sleep by providing integrated wellness insights that combine movement, calorie expenditure, and sleep metrics into a single daily wellness summary.

Expected Business Value

Promoting a holistic approach to health aligns with Bellabeat's mission and helps users better understand the relationship between different aspects of their wellness.

Recommendation 4: Deliver Personalized Sleep Insights

Supporting Finding

Users spent an average of **39 minutes awake while in bed**, indicating a difference between time spent in bed and actual sleep duration.

Recommendation

Bellabeat should expand its sleep reporting by providing users with insights such as sleep efficiency, bedtime consistency, and personalized recommendations for improving sleep quality.

Expected Business Value

Enhanced sleep analytics can increase the perceived value of Bellabeat's products while encouraging continued user engagement.

Overall Recommendation

The analysis indicates that Fitbit users actively monitor both physical activity and sleep, making these two areas central to Bellabeat's value proposition. Bellabeat should focus its marketing strategy on promoting a balanced approach to wellness by encouraging consistent daily activity, supporting healthy sleep habits, and delivering personalized health insights based on user behavior. By combining these features with targeted engagement strategies, Bellabeat can strengthen user retention, enhance the customer experience, and reinforce its position as a health-focused smart device company.

7. CONCLUSION

The objective of this case study was to analyze Fitbit smart device usage data to identify consumer behavior patterns and generate insights that could support Bellabeat's marketing strategy.

The analysis revealed that Fitbit users maintained moderate levels of daily physical activity while actively tracking both exercise and sleep. Users averaged **7,638 daily steps** and **2,304 calories burned**, with activity levels peaking on **Saturdays** and **Tuesdays**. Sleep analysis showed that users averaged approximately **seven hours of sleep** per night, while spending about **39 minutes awake in bed**.

Further analysis demonstrated a strong positive relationship between daily step count and calorie expenditure, indicating that users who recorded more daily steps generally burned more calories. However, higher activity levels were associated with shorter average sleep durations within this dataset, highlighting the importance of considering multiple aspects of health rather than relying on a single wellness metric.

Based on these findings, recommendations were developed to encourage consistent physical activity, promote a balanced approach to wellness, and enhance personalized health insights for Bellabeat users. While the dataset has limitations, including a relatively small sample size, the absence of demographic information, and data collected in 2016, the analysis provides valuable insights into smart device usage patterns.